

SERO Emergency Response Optimization: Example Workflow

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1 Introduction

This document demonstrates a typical user workflow for emergency location optimization using the SERO package. It covers data loading, accident visualization, optimal location finding, and professional mapping, with all plots and results shown inline.

2 1. Load Data

```
library(SERO)
library(sf)
```

```
## Linking to GEOS 3.13.0, GDAL 3.8.5, PROJ 9.5.1; sf_use_s2() is TRUE
```

```

library(ggplot2)
# Load demo data (replace with your own data as needed)
data <- sero_load_data()
str(data)

## List of 5
## $ accident :Classes 'sf' and 'data.frame': 5695 obs. of 25 variables:
## ..$ ULAND : int [1:5695] 5 5 5 5 5 5 5 5 5 5 ...
## ..$ UREGBEZ : int [1:5695] 5 5 5 5 5 5 5 5 5 5 ...
## ..$ UKREIS : int [1:5695] 15 15 15 15 15 15 15 15 15 15 ...
## ..$ UGEMEINDE : int [1:5695] 0 0 0 0 0 0 0 0 0 0 ...
## ..$ UJAHR : int [1:5695] 2019 2019 2019 2019 2019 2019 2019 2019 2019 ...
## ..$ UMONAT : int [1:5695] 1 1 1 1 1 1 1 1 1 1 ...
## ..$ USTUNDE : int [1:5695] 20 18 17 13 6 13 5 11 9 18 ...
## ..$ UWOCHENTAG : int [1:5695] 3 4 4 4 4 6 3 5 2 2 ...
## ..$ UKATEGORIE : int [1:5695] 2 3 2 2 3 3 3 3 2 3 ...
## ..$ UART : int [1:5695] 5 2 4 3 8 5 0 2 6 6 ...
## ..$ UTYP1 : int [1:5695] 3 6 7 6 1 2 7 6 4 4 ...
## ..$ ULICHTVERH : int [1:5695] 2 2 2 0 2 0 2 0 0 2 ...
## ..$ IstRad : int [1:5695] 0 0 0 0 0 1 1 0 0 0 ...
## ..$ IstPKW : int [1:5695] 1 1 1 1 1 1 0 1 1 1 ...
## ..$ IstFuss : int [1:5695] 0 0 0 0 0 0 0 0 1 1 ...
## ..$ IstKrad : int [1:5695] 0 0 0 0 0 0 0 0 0 0 ...
## ..$ IstGkfz : int [1:5695] 0 0 0 0 0 0 0 0 0 0 ...
## ..$ IstSonstige: int [1:5695] 0 0 0 0 0 0 0 0 0 0 ...
## ..$ LINREFX : num [1:5695] 411665 408200 410687 406835 409673 ...
## ..$ LINREFY : num [1:5695] 5751946 5754184 5755399 5756778 5753889 ...
## ..$ XGCSWGS84 : num [1:5695] 7.72 7.66 7.7 7.64 7.69 ...
## ..$ YGCSWGS84 : num [1:5695] 51.9 51.9 51.9 52 51.9 ...
## ..$ STRZUSTAND : int [1:5695] 1 0 0 0 2 0 2 1 1 1 ...
## ..$ UIDENTSTLAE: num [1:5695] NA NA NA NA NA NA NA NA NA NA ...
## ..$ geom :sfc_POINT of length 5695; first list element: 'XY' num [1:2] 7.72 51.91
## ..- attr(*, "sf_column")= chr "geom"
## ..- attr(*, "agr")= Factor w/ 3 levels "constant","aggregate",...: NA NA NA NA NA NA NA NA NA NA ...
## ..- attr(*, "names")= chr [1:24] "ULAND" "UREGBEZ" "UKREIS" "UGEMEINDE" ...
## $ landuse :Classes 'sf' and 'data.frame': 13284 obs. of 5 variables:
## ..$ osm_id: chr [1:13284] "24483473" "24744788" "25200097" "25640695" ...
## ..$ code : int [1:13284] 7211 7201 7206 7209 7206 7203 7201 7218 7228 7228 ...
## ..$ fclass: chr [1:13284] "recreation_ground" "forest" "cemetery" "commercial" ...
## ..$ name : chr [1:13284] "Freibad Stapelskotten" NA "Friedhof Sankt Clemens" "Halle Münsterland"
## ..$ geom :sfc_MULTIPOLYGON of length 13284; first list element: List of 1
## ..- attr(*, "sf_column")= chr "geom"
## ..- attr(*, "agr")= Factor w/ 3 levels "constant","aggregate",...: NA NA NA NA
## ..- attr(*, "names")= chr [1:4] "osm_id" "code" "fclass" "name"
## $ population:Classes 'sf' and 'data.frame': 46 obs. of 7 variables:
## ..$ name : chr [1:46] "Aaseestadt" "Aegidii" "Albachten" "Amelsbüren" ...
## ..$ population: int [1:46] 6045 1441 1402 2357 2505 1306 4649 8752 1705 7269 ...
## ..$ area : num [1:46] 1.462 0.164 12.742 43.528 5.088 ...
## ..$ density : num [1:46] 4133.6 8799.2 110 54.1 492.4 ...
## ..$ layer : chr [1:46] "1" "population_density" "population_density" "population_density" ...

```

```
## ..$ path      : chr [1:46] "1.geojson" "C:/Users/User/Desktop/IFGI/SII/project/data/outputsii12.ge
## ..$ geom      :sfc_POLYGON of length 46; first list element: List of 1
## .. ..$ : num [1:108, 1:2] 7.59 7.6 7.6 7.6 7.6 ...
## .. ..- attr(*, "class")= chr [1:3] "XY" "POLYGON" "sfg"
## ..- attr(*, "sf_column")= chr "geom"
## ..- attr(*, "agr")= Factor w/ 3 levels "constant","aggregate",...: NA NA NA NA NA NA
## .. ..- attr(*, "names")= chr [1:6] "name" "population" "area" "density" ...
## $ roads       :Classes 'sf' and 'data.frame': 40175 obs. of  11 variables:
## ..$ osm_id    : chr [1:40175] "5183767" "5608019" "5608027" "5736734" ...
## ..$ code      : int [1:40175] 5112 5152 5153 5152 5152 5114 5114 5122 5122 5114 ...
## ..$ fclass    : chr [1:40175] "trunk" "cycleway" "footway" "cycleway" ...
## ..$ name      : chr [1:40175] "Umgehungsstraße" "Hohenzollernring" "Wolbecker Straße" NA ...
## ..$ ref       : chr [1:40175] "B 51" NA NA NA ...
## ..$ oneway    : chr [1:40175] "F" "F" "F" "F" ...
## ..$ maxspeed  : int [1:40175] 70 0 0 0 0 50 50 30 30 50 ...
## ..$ layer     : num [1:40175] -1 0 0 0 0 0 0 0 0 0 ...
## ..$ bridge    : chr [1:40175] "F" "F" "F" "F" ...
## ..$ tunnel    : chr [1:40175] "T" "F" "F" "F" ...
## ..$ geom      :sfc_MULTILINESTRING of length 40175; first list element: List of 1
## .. ..$ : num [1:2, 1:2] 7.65 7.65 51.94 51.94
## .. ..- attr(*, "class")= chr [1:3] "XY" "MULTILINESTRING" "sfg"
## ..- attr(*, "sf_column")= chr "geom"
## ..- attr(*, "agr")= Factor w/ 3 levels "constant","aggregate",...: NA NA NA NA NA NA NA NA NA
## .. ..- attr(*, "names")= chr [1:10] "osm_id" "code" "fclass" "name" ...
## $ districts   :Classes 'sf' and 'data.frame': 6 obs. of  8 variables:
## ..$ OBJECTID  : num [1:6] 1 2 3 4 5 6
## ..$ STADTBZIR: chr [1:6] "8 Münster-Südost" "9 Münster-Hiltrup" "7 Münster-Ost" "6 Münster-Nord"
## ..$ NR_STADTBE: chr [1:6] "8" "9" "7" "6" ...
## ..$ NAME_STADT: chr [1:6] "Münster-Südost" "Münster-Hiltrup" "Münster-Ost" "Münster-Nord" ...
## ..$ STAND     : Date[1:6], format: "2018-12-18" "2018-12-18" ...
## ..$ SHAPE_AREA: num [1:6] 36116459 69798033 54196963 36067725 83005889 ...
## ..$ SHAPE_LEN : num [1:6] 35273 49462 41458 32113 53563 ...
## ..$ geom      :sfc_MULTIPOLYGON of length 6; first list element: List of 1
## .. ..$ :List of 1
## .. .. ..$ : num [1:797, 1:2] 412917 412909 412908 412598 412036 ...
## .. ..- attr(*, "class")= chr [1:3] "XY" "MULTIPOLYGON" "sfg"
## ..- attr(*, "sf_column")= chr "geom"
## ..- attr(*, "agr")= Factor w/ 3 levels "constant","aggregate",...: NA NA NA NA NA NA NA
## .. ..- attr(*, "names")= chr [1:7] "OBJECTID" "STADTBZIR" "NR_STADTBE" "NAME_STADT" ...
```

3 2. Visualize Accidents by Severity

```
# Plot accidents, focusing on fatal and serious categories
accident_plot <- sero_plot_accidents(
  accidents = data$accident,
  districts = data$districts,
  landuse = data$landuse,
  severity_levels = c("fatal", "serious"),
  district_border_color = "navy",
  accident_color = "red",
  accident_shape = 19,
```

```

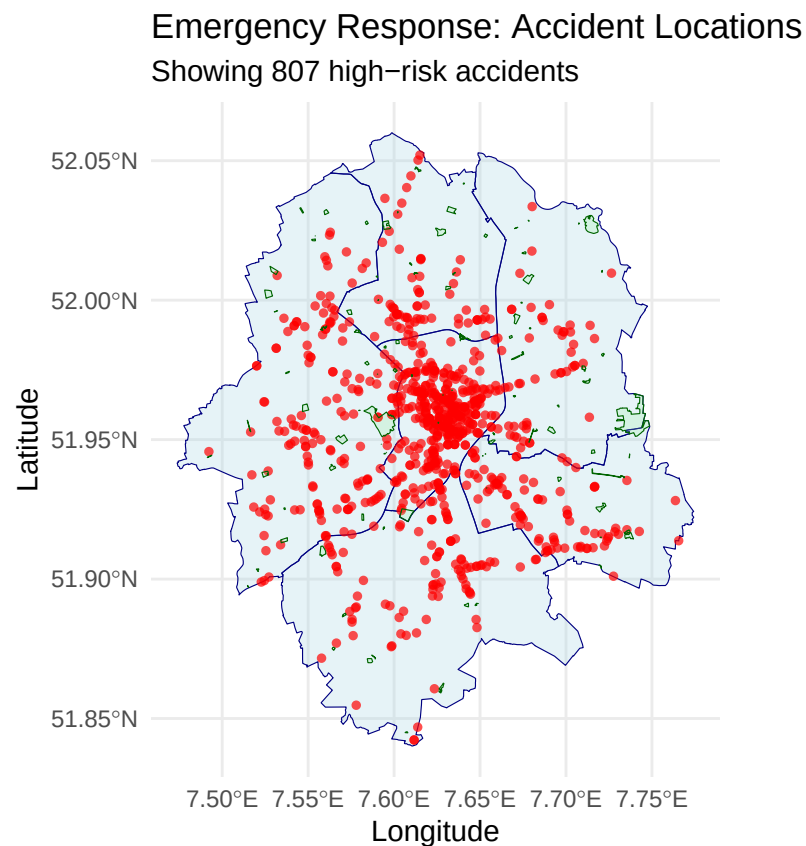
landuse_border_color = "darkgreen",
landuse_fill_color = "lightgreen",
landuse_sample_size = 100
)

```

```
## Filtered to 807 accidents with severity levels: 1, 2
```

```
## Sampled 100 land use polygons for performance.
```

```
accident_plot
```



4 3. Find Optimal Emergency Locations

```

optimal_locs <- sero_find_optimal_locations(
  data = data,
  num_locations = 5,
  method = "fast",
  risk_category_names = c("fatal", "serious"),
  quick = TRUE
)

```

```
## Using 807 high-risk accidents (categories 1, 2 )
## Found 5 optimal emergency service locations
```

```
print(optimal_locs)
```

```
## Simple feature collection with 5 features and 12 fields
## Geometry type: POINT
## Dimension: XY
## Bounding box: xmin: 7.556272 ymin: 51.90893 xmax: 7.692374 ymax: 51.99536
## Geodetic CRS: WGS 84
##   location_id optimization_method OBJECTID      STADTBEZIR NR_STADTBE
## 1    BASE_1          k-means         4      6 Münster-Nord         6
## 2    BASE_2          k-means         5      5 Münster-West         5
## 3    BASE_3          k-means        2 9 Münster-Hiltrup         9
## 4    BASE_4          k-means         1  8 Münster-Südost         8
## 5    BASE_5          k-means         6      1 Mitte              1
##   NAME_STADT      STAND SHAPE_AREA SHAPE_LEN      geometry
## 1 Münster-Nord 2018-12-18  36067725 32113.18 POINT (7.584182 51.99536)
## 2 Münster-West 2018-12-18  83005889 53562.74 POINT (7.556272 51.92994)
## 3 Münster-Hiltrup 2018-12-18  69798033 49462.06 POINT (7.625012 51.90893)
## 4 Münster-Südost 2018-12-18  36116459 35272.85 POINT (7.692374 51.94037)
## 5      Mitte 2018-12-18  23935274 23377.15 POINT (7.630538 51.96194)
##   accident_count_500m accident_count_1000m accessibility_score
## 1                   0                      5                0.292
## 2                   4                      11                0.751
## 3                   5                      14                0.879
## 4                   0                       3                0.219
## 5                  37                     119                8.904
```

5 4. Visualize Optimal Locations

```
optimal_plot <- sero_plot_optimal_locations(
  optimal_locs = optimal_locs,
  districts = data$districts,
  roads = data$roads,
  landuse = data$landuse,
  accidents = data$accident,
  district_border_color = "black",
  road_border_color = "gray",
  landuse_border_color = "darkgreen",
  landuse_fill_color = "lightgreen",
  landuse_sample_size = 100,
  road_sample_size = 200,
  optimal_color = "red",
  optimal_shape = 17
)
```

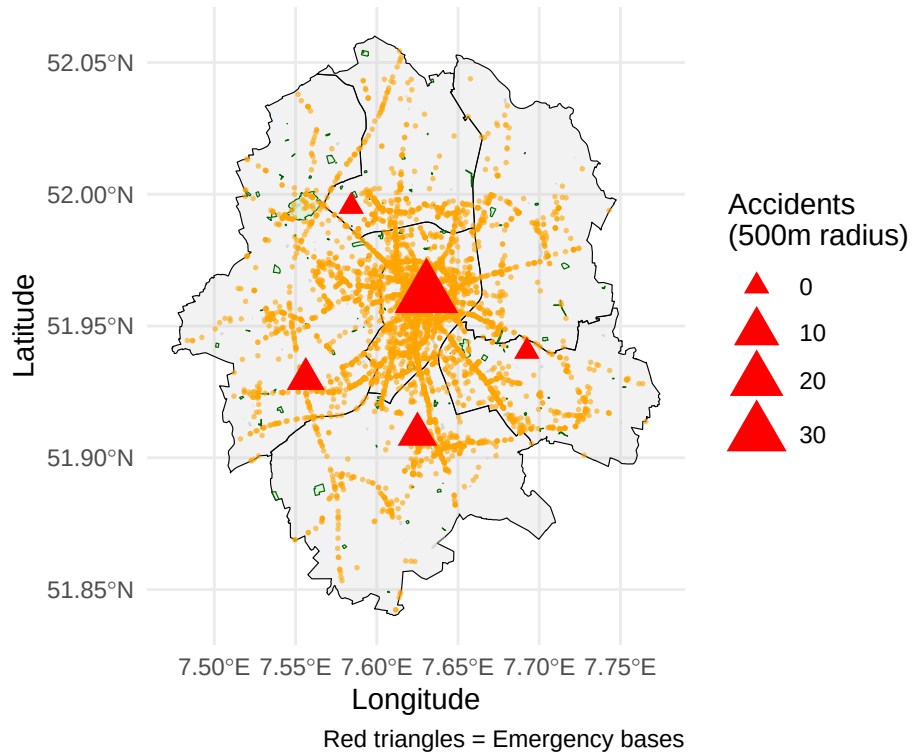
```
## Sampled 200 road segments for performance.
```

```
## Sampled 100 land use polygons for performance.
```

```
optimal_plot
```

Optimal Emergency Service Locations

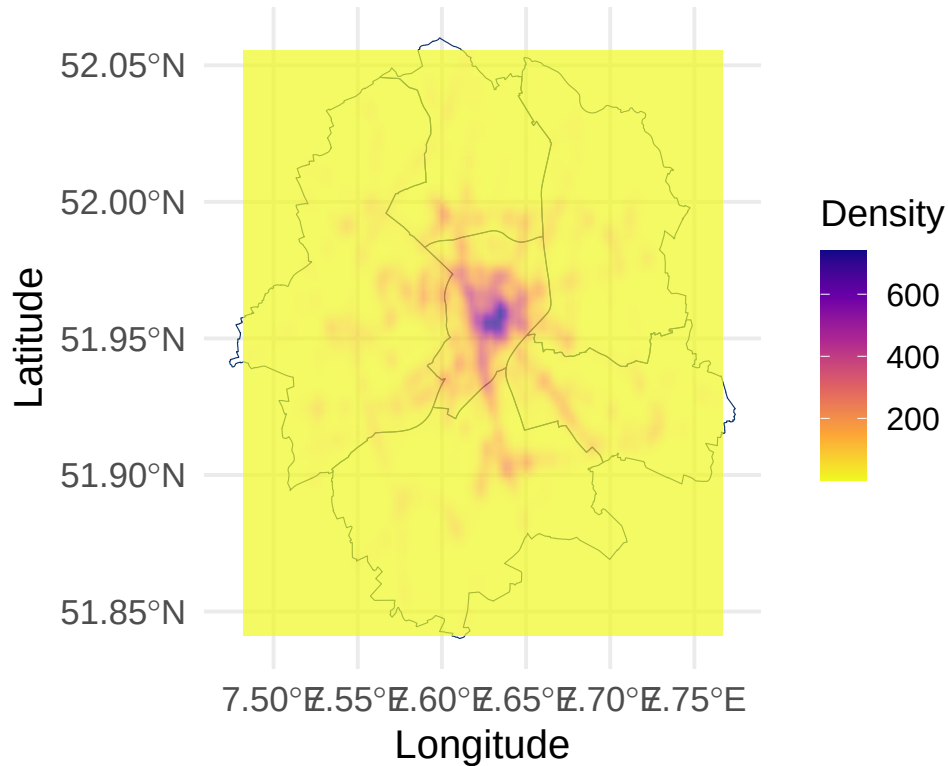
Found 5 strategic locations



6 5. Create Accident Hotspots (Kernel Density)

```
hotspot_plot <- sero_hotspots(data$accident, districts = data$districts, bandwidth = 0.01)
hotspot_plot
```

Accident Hotspots (Kernel Density)

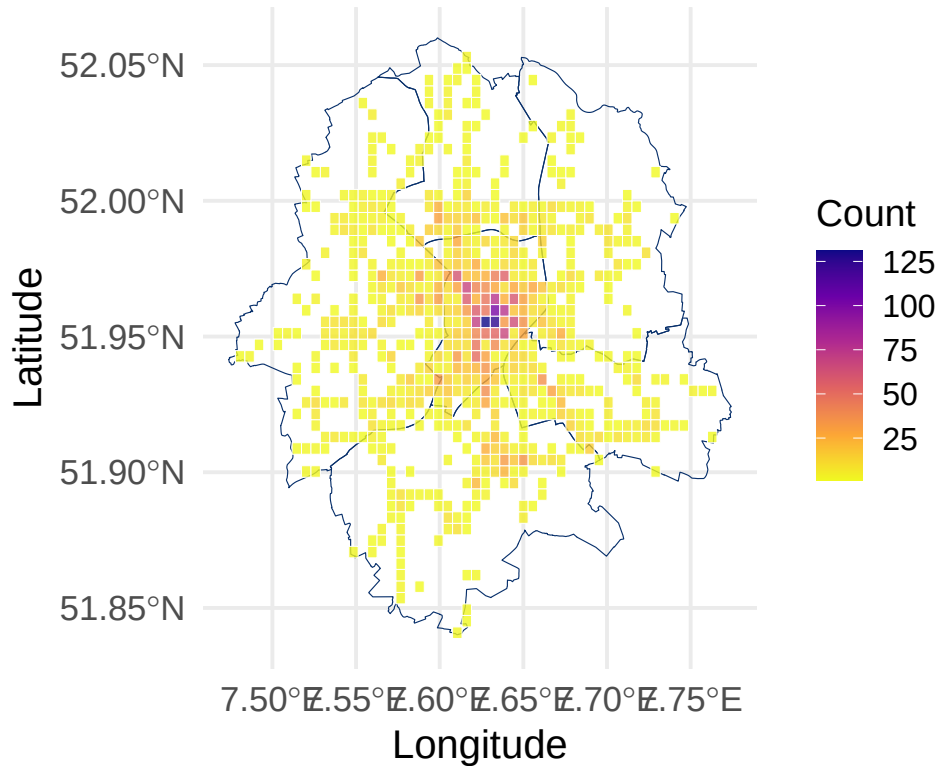


7 6. Create Accident Heatmap (2D Histogram)

```
heatmap_plot <- sero_heatmap(data$accident, districts = data$districts, bins = 50)
heatmap_plot
```

```
## Warning: The dot-dot notation ('..count..') was deprecated in ggplot2 3.4.0.
## i Please use 'after_stat(count)' instead.
## i The deprecated feature was likely used in the SERO package.
## Please report the issue to the authors.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was generated.
```

Accident Heatmap (2D Histogram)



8 7. Performance Summary

```
cat("Emergency bases found:", nrow(optimal_locs), "\n")
```

```
## Emergency bases found: 5
```

```
if ("accessibility_score" %in% names(optimal_locs)) {
  cat("Average accessibility score:", round(mean(optimal_locs$accessibility_score), 3), "\n")
  cat("Best accessibility score:", round(max(optimal_locs$accessibility_score), 3), "\n")
}
```

```
## Average accessibility score: 2.209
```

```
## Best accessibility score: 8.904
```

```
if ("accident_count_500m" %in% names(optimal_locs)) {
  cat("Average accidents covered (500m):", round(mean(optimal_locs$accident_count_500m), 1), "\n")
  cat("Maximum accidents covered (500m):", max(optimal_locs$accident_count_500m), "\n")
}
```

```
## Average accidents covered (500m): 9.2
```

```
## Maximum accidents covered (500m): 37
```


9 8. Save Results

```
sero_save_optimal_locations(optimal_locs, "optimal_locations.sqlite")
```

```
## Optimal locations saved to database: optimal_locations.sqlite
```

10 9. Session Info

```
sessionInfo()
```

```
## R version 4.5.1 (2025-06-13)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.0
##
## Matrix products: default
## BLAS: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; LAPACK v
##
## locale:
## [1] C.UTF-8/C.UTF-8/C.UTF-8/C/C.UTF-8/C.UTF-8
##
## time zone: Europe/Berlin
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods    base
##
## other attached packages:
## [1] ggplot2_3.5.2 sf_1.0-21      SERO_0.1.0
##
## loaded via a namespace (and not attached):
## [1] bit_4.6.0          gtable_0.3.6      crayon_1.5.3      dplyr_1.1.4
## [5] compiler_4.5.1     tidyselect_1.2.1  Rcpp_1.1.0        blob_1.2.4
## [9] dichromat_2.0-0.1  scales_1.4.0      yaml_2.3.10       fastmap_1.2.0
## [13] R6_2.6.1           labeling_0.4.3    generics_0.1.4    classInt_0.4-11
## [17] s2_1.1.9           knitr_1.50        MASS_7.3-65       tibble_3.3.0
## [21] units_0.8-7        DBI_1.2.3         pillar_1.11.0     RColorBrewer_1.1-3
## [25] rlang_1.1.6        cachem_1.1.0      xfun_0.52         bit64_4.6.0-1
## [29] memoise_2.0.1      RSQLite_2.4.1     viridisLite_0.4.2 cli_3.6.5
## [33] withr_3.0.2        magrittr_2.0.3    class_7.3-23      wk_0.9.4
## [37] digest_0.6.37      grid_4.5.1        lifecycle_1.0.4   vctrs_0.6.5
## [41] KernSmooth_2.23-26 proxy_0.4-27       evaluate_1.0.4    glue_1.8.0
## [45] farver_2.1.2       e1071_1.7-16      rmarkdown_2.29    tools_4.5.1
## [49] pkgconfig_2.0.3    htmltools_0.5.8.1
```